



ColdDTA: Utilizing data augmentation and attention-based feature fusion for drug-target binding affinity prediction

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ABSTRACT

Accurate prediction of drug-target affinity (DTA) plays a crucial role in drug discovery and development. Recently, deep learning methods have shown excellent predictive performance on randomly split public datasets. However, verifications are still required on this splitting method to reflect real-world problems in practical applications. And in a cold-start experimental setup, where drugs or proteins in the test set do not appear in the training set, the performance of deep learning models often significantly decreases. This indicates that improving the generalization ability of the models remains a challenge. To this end, in this study, we propose ColdDTA: using data augmentation and attention-based feature fusion to improve the generalization ability of predicting drug-target binding affinity. Specifically, ColdDTA generates new drug-target pairs by removing subgraphs of drugs. The attention-based feature fusion module is also used to better capture the drug-target interactions. We conduct cold-start experiments on three benchmark datasets, and the consistency index (CI) and mean square error (MSE) results on the Davis and KIBA datasets show that ColdDTA outperforms the five state-of-the-art baseline methods. Meanwhile, the results of area under the receiver operating characteristic (ROC-AUC) on the BindingDB dataset show that ColdDTA also has better performance on the classification task. Furthermore, visualizing the model weights allows for interpretable insights. Overall, ColdDTA can better solve the realistic DTA prediction problem. The code has been available to the public.

1. Introduction

Drug development is a lengthy and expensive process. According to the survey, it takes around 13 years and costs 0.6–2.6 billion dollars to develop a new drug, with a success rate of less than 10% [1]. This has prompted the use of computational methods to reduce costs and increase efficiency. The process of drug molecules binding to targets with appropriate chemical characteristics and affinity is one of the key criteria for curing diseases. Therefore, making accurate predictions of drug-target binding affinity is essential in drug discovery [2].

Methods for drug-target affinity (DTA) prediction using computers can be classified into structure-based approaches, ligand-based approaches, and hybrid approaches [3]. Molecular docking is one of the

earliest and most widely used structure-based approaches. However, this approach requires the 3D structure of the protein and a long computational time, even now, there are still a large number of unknown protein structures. Although AlphaFold2 is able to predict the majority of proteins [4], not all results are highly accurate [5]. Ligand-based approaches assume that drug molecules with similar chemical properties have the same chemical activity therefore can interact with similar targets. Nevertheless, when the number of known ligands is insufficient, the prediction results of ligand-based methods often become unreliable. The hybrid approaches are expected to overcome the limitations stated above. One of them, traditional machine learning methods such as Support Vector Machine (SVM) and Random Forest (RF) require feature engineering to extract drug and protein

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features [6], feature engineering is time-consuming and can affect prediction performance due to information loss. In recent years, several deep learning methods, like DeepDTA [7] and GraphDTA [8], have been developed for drug-target interaction prediction.

Some deep learning models use sequence-based input data, such as WideDTA [9], to represent drugs with Simplified Molecular Input Line Entry System (SMILES) and represent proteins with amino acid sequences, and then extract features through Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs). But such methods lack molecular topology. To compensate for this deficiency, many methods regard the drugs as graph structure, with the atoms and chemical bonds of the compound corresponding to the vertices and edges of the graph, and then perform feature extraction by Graph Neural Networks (GNNs). DeepGS [10], MolTrans [11], and MgraphDTA [12] all represent drugs as graphs and showed excellent prediction performance on some public datasets. Similarly, several studies have focused on the role played by proteins in the drug-target binding process, where DrugVQA [13], AttentionSiteDTI [14], and GSAML-DTA [15] used contact maps to represent proteins, and experimental results showed that better predictions can be made by using the 3D structure of the protein. Transformer [16] has also been applied to predict drug-target interactions. The TransformerCPI model proposed by Chen et al. treats drug-target interactions as a binary classification problem and uses an encoder-decoder architecture for model training, ultimately predicting whether the two can interact with each other [17].

Although some deep learning methods exhibit excellent predictive performance on DTA problems, these methods are mostly evaluated on randomly split datasets (where drugs and targets in the test set have already appeared in the training set), which can cause information leakage and make the results overly optimistic [18]. From an application perspective, most proteins or drugs do not appear in the training set [19]. When the dataset is split according to more realistic scenarios, the predictive performance of many models drops dramatically [20], and sometimes the predictive performance is worse than that of traditional machine learning methods [21]. This indicates that the generalization ability of current DTA models still needs to be improved.

In order to improve the generalization ability of the model and enable it to play a greater role in real scenarios, we first considered the use of data augmentation methods. Data augmentation methods have been widely used in computer vision [22–24] and natural language processing [25–27], and have been shown to effectively improve model generalization. Multiple effective data augmentation methods have also been proposed for predicting molecular properties [28,29]. However, to the best of our knowledge, data augmentation has not been used to improve the generalization performance of DTA prediction methods that represent drugs using molecular graphs. This may be due to the fact that biochemistry tells us that small changes in atoms in a compound may also lead to significant changes in the physical and chemical properties of the molecule, as well as changes in the binding affinity between the small drug molecule and the target. As shown in Fig. 1, for a catechol molecule, deleting some atoms to a phenol changes its water solubility from soluble in water to slightly soluble in water. On the other hand, feature fusion refers to the combination of features from different layers

or branches and is ubiquitous in modern network architectures, and the use of reasonable feature fusion methods can effectively improve model performance. But most existing DTA prediction methods simply concatenate the features of drugs and proteins and perform final result prediction, which limits the interaction between drugs and targets and restricts the predictive and generalization performance of the model.

In this work, we present a drug-target affinity training method called ColdDTA, which uses data augmentation and attention-based feature fusion to predict the binding affinity between drugs and proteins. Specifically, a certain percentage of subgraphs of the drug is removed and constitutes new data pairs with the original target in the data processing phase, then used in the training phase. Moreover, ColdDTA uses a drug-protein feature fusion module to replace the simple concatenation of features. By conducting experiments on two benchmark datasets, we find that ColdDTA achieved the best overall performance compared to other methods. The results of our ablation experiments also demonstrate the effectiveness of the data augmentation method and the drug-target fusion module. The main contributions of this study are summarized as follows:

1. A new data augmentation method applicable to the DTA task is used by removing subgraphs of drugs and forming new data pairs with the original targets, which can effectively improve the generalization ability of the model.
2. An attention-based data fusion method between drug and target is proposed, which can better integrate drug features with protein features and benefit the model for affinity value prediction.
3. We conducted extensive experiments on two public datasets and verified that ColdDTA has a significant improvement in prediction ability compared to the baseline model under the dataset setting close to the real experimental environment. Also, the experiments demonstrate the effectiveness of data augmentation and drug-target fusion.

2. Materials and methodology

2.1. Overview

We propose an end-to-end training method called ColdDTA to improve DTA prediction performance. We define the DTA problem as follows: Let $X = ((d_b, p_b, y_b); b \in (1, \dots, B))$ be a batch of labeled data, where d is the SMILES representation of a small molecule drug, p is the protein, and y is the binding affinity value obtained through experiments. The overall training process of the model is shown in Fig. 2. It consists of five parts: data augmentation module, drug feature extraction module, protein feature extraction module, drug-target fusion module, and prediction module.

2.2. Data augmentation strategy

Data augmentation is a necessary condition for improving data efficiency and enhancing the generalization performance of neural networks [30]. It seems obvious that using data augmentation methods can

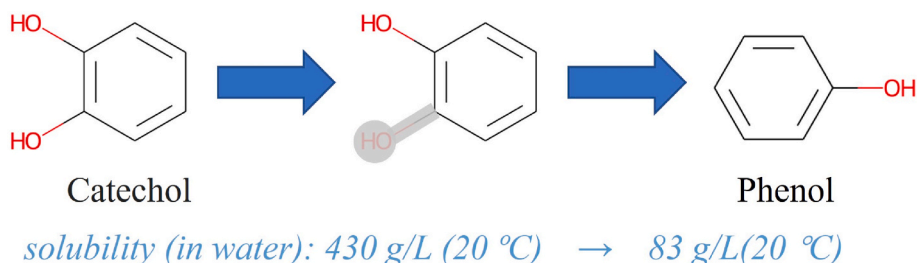


Fig. 1. Illustration of changes in atoms that affect its properties.

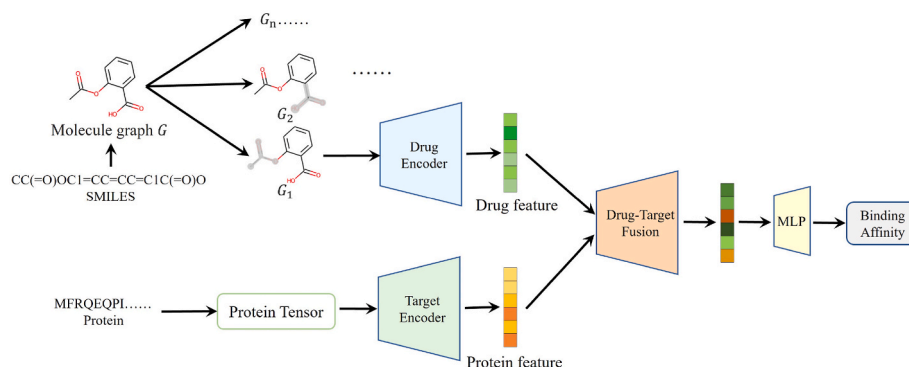


Fig. 2. The framework of the proposed ColdDTA. Firstly, the subgraph of the drug is removed and new pairs are formed with the original protein. Second, the features of the molecular graph and the protein are respectively extracted using GNN and CNN. Third, the features from the previous step are input to the drug-target fusion module for interaction to get a more efficient feature representation. Finally, the fused feature is fed into a MLP to predict binding affinity.

improve model predictive performance, but rare studies are reported using this strategy for the drug-target interaction prediction problem, so we introduced a novel data augmentation strategy in DTA prediction. Specifically, in drug-target pairing, we first randomly select an atom in the molecular graph as the starting node, then remove the starting node and recursively remove its neighboring nodes until a certain percentage

of the subgraph is removed, while the corresponding target and binding affinity remain unchanged. In this process, the removal of atomic groups is completely random, which is similar to the contrastive learning for molecular property prediction [31]. Fig. 3 shows the output of the data augmentation process. In the training phase, a pair of a drug compound and a target protein generates multiple new pairs by removing

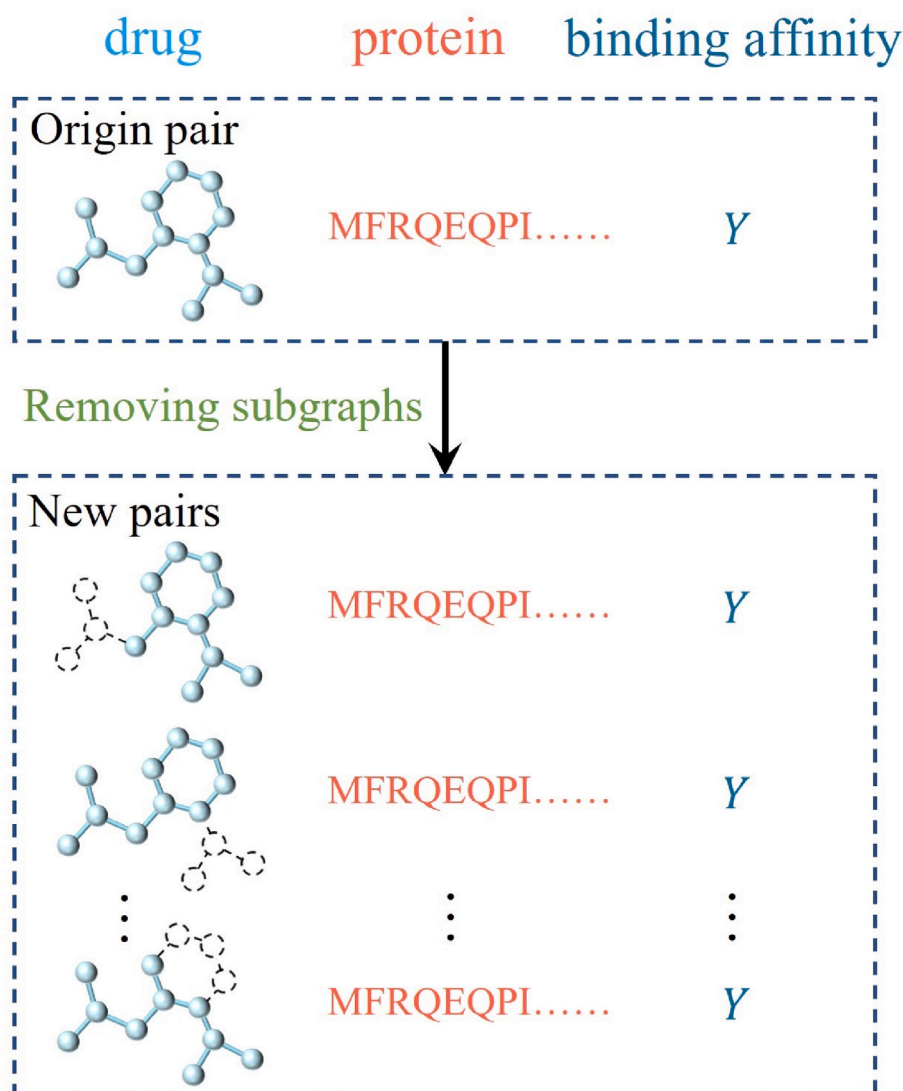


Fig. 3. The output of the data augmentation process.

subgraphs. It is important to note that no subgraphs are removed during the testing phase.

2.3. Drug feature extraction module

The drug features $y_G \in R^m$ are extracted using GNN. Compounds are represented using molecular graphs G , defined as $G = (V, E)$, where V is the set of vertices in the graph, and E is the set of edges. The atomic type, atomic hybridization mode, total number of hydrogen atoms, and atomic valence are used to compute the feature vector for each atom. The features of each vertex are obtained by aggregating and transferring features from its neighboring vertices through multiple rounds. The message-passing phase can be formulated as follows:

$$h_i^{(k+1)} = U_k \left((1 + \omega^{(k)}) h_i^{(k)} + \sum_{j \in N(i)} h_j^{(k)} \right) \quad (1)$$

where $h_i^{(k)} \in R^m$ represents the feature vector of vertex i at time step k , $N(i)$ is the set of neighboring vertices of vertex i , U_k is a vertex update function, and ω is a learnable parameter. This updates the feature of vertex i at the next time step. To calculate the feature vector of the entire graph, we use a readout function:

$$y_G = \frac{1}{|V|} \sum_{v \in V} h_v^{(k)} \quad (2)$$

where $|V|$ is the number of vertices in the molecular graph, and $h_v^{(k)}$ is the feature vector of vertex v at time step k .

In our experiments, residual connections [32] were used between each GNN layer to prevent information loss.

2.4. Protein feature extraction module

Proteins are represented by amino acid sequences, and different amino acids are encoded using different numbers, such as alanine (A) is represented by the integer 1, cysteine (C) is 2, aspartic acid (D) is 5, and so on, to obtain an integer sequence that can represent the protein. Following previous work [33], the amino acid sequence length of the protein is set to 1200 to facilitate model training, and the portion of the amino acid sequence that exceeds the length is directly truncated, and the insufficient portion is supplemented with filled zeros. Then, each integer representing an amino acid is mapped to a learnable 128-dimensional vector by an embedding layer, where each integer represents a unique vector. At this point, the input matrix $S \in R^{n \times e}$ is obtained, where n is the maximum length of the amino acid sequence and e is the embedding size of the protein. After that, we extract the global features of the sequence using three 1D-convolutional layers. The features obtained from each layer can be expressed as:

$$X_p^{(l+1)} = B \circ \delta \circ F(X_p^{(l)}) \quad (3)$$

where $X_p^{(l)}$ is the l -th hidden protein representation and $X_p^{(0)} = S$. F denotes a CNN layer, δ denotes the Rectified Linear Unit (ReLU), and B denotes the Batch Normalization (BN). $B \circ \delta \circ F$ is a composite function of B and δ and F , which means that operations F , δ , B are performed sequentially on the representations.

2.5. Drug-target fusion module

From a biological perspective, the binding of drugs and targets is often performed through the lock-and-key principle, whereby the protein acts as a lock that needs to be opened, and the drug acts as the key that needs to match in order for binding to take place [34]. The basic principle of molecular docking is the lock-and-key model [35]. But most existing end-to-end models simply concatenate the features of drugs and proteins and then use the MLP to adaptively capture the relationship

between them. This makes almost no connection between drugs and proteins, and it ignores the actual interaction between drugs and targets in reality, which limits the predictive performance and generalization ability of the model. And from a model perspective, in DTA prediction tasks, larger models are often used to extract features from drugs, while smaller models are used to extract features from proteins. The features of drugs and proteins are not equally important in the interaction prediction stage.

To better capture the interaction mechanism between drugs and proteins, we designed a feature fusion module based on attention mechanisms, called drug-target fusion, which is illustrated in Fig. 4. Given a drug feature $f_d \in R^d$ and a protein feature $f_p \in R^d$, which have the same dimension, denoted by d . First, the attention weights $W_1 \in R^d$ can be computed with the following equation:

$$W_1 = \sigma \circ B \circ \tau \circ \delta \circ B \circ \tau (f_d \oplus f_p) \quad (4)$$

where $\oplus$ denotes element-wise addition, $\circ$ takes the output of the right function as the input of the left function, τ denotes a linear layer, and σ

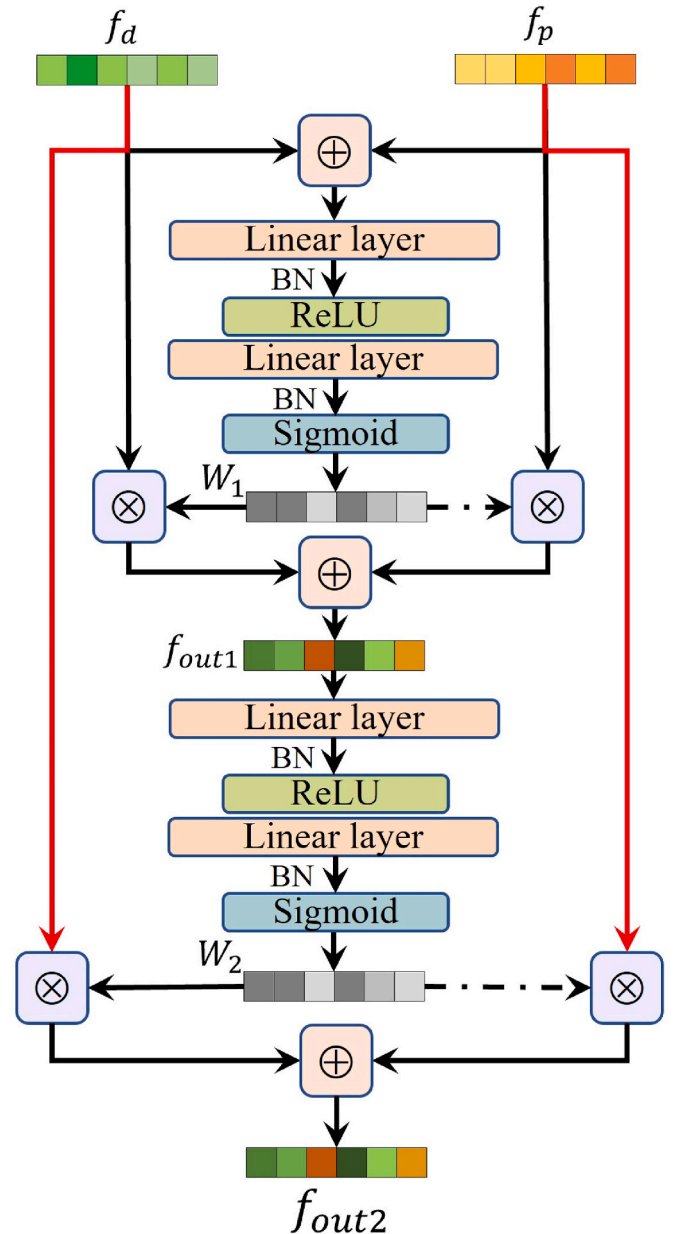


Fig. 4. Illustration of the proposed Drug-target fusion.

denotes the sigmoid function. Then, the feature $f_{out1} \in R^d$ after the first fusion can be computed as follows:

$$f_{out1} = W_1 * f_d + (1 - W_1) * f_p \quad (5)$$

However, the quality of the initial features of drugs and proteins may affect the final fusion weight. Another attention module is used for feature fusion to obtain a new fusion weight $W_2 \in R^d$:

$$W_2 = \sigma(B \circ \tau \circ \delta \circ B \circ \tau)(f_{out1}) \quad (6)$$

$$f_{out2} = W_2 * f_d + (1 - W_2) * f_p \quad (7)$$

The resulting $f_{out2} \in R^d$ is the final output of the feature fusion. This method of feature fusion based on attention mechanisms provide a way to effectively capture the information of drug-target interactions, thus making the model more accurate in DTA prediction and improving its generalization ability.

2.6. Prediction module

The feature tensor after drug-target fusion is passed through a classifier containing multiple fully connected layers for the prediction of the final result. We use four fully connected layers with the number of neurons 1024,1024,256 and 1. Each linear layer is connected by a BatchNormalization layer and a ReLU activation function. Meanwhile, the loss function used in the training process of the model is MSELoss:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (8)$$

where $\hat{y}_i$ is the predicted value of the i -th drug-target pair, y_i is true value, and n is the number of samples.

We simply replaced the MSE loss with a cross-entropy loss when the DTA prediction was treated as a binary classification problem.

2.7. Datasets

We conducted experiments on three public datasets and compared the experimental results with other advanced models to validate the merits of ColdDTA. On the regression task, the datasets used for the experiments are Davis dataset [36] and KIBA dataset [37]. Also, we used the BindingDB dataset [38] for the classification problem. The information of the datasets can be found in Table 1.

Davis dataset was obtained from the human catalytic protein kinase group tested by Davis et al. The dataset has 68 drugs and 442 proteins with a total of 30056 pairs of interaction relationships, and the binding affinity between the drug and the target is obtained in the wet lab by determining the dissociation constant (K_d) value, and for better training of the model, the K_d values obtained in the lab are transformed as follows:

$$pK_d = -\log_{10}(K_d/10^9) \quad (9)$$

KIBA dataset contains 2111 drugs and 229 proteins with a total of 118254 interaction pairs. The values of the two interactions were expressed by KIBA score, and the KIBA score was used to optimize the consistency of dissociation constant K_d , inhibition constant K_i , and IC_{50} .

BindingDB dataset contains experimentally based assays of interactions between small molecules and proteins. In total, the dataset consists of K_d values among 10665 drugs and 1413 proteins. We labeled

Table 1
Statistics of the three datasets.

Datasets	Task type	Drugs	Proteins	Interactions
Davis	Regression	68	442	30056
KIBA	Regression	2111	229	118254
BindingDB	Classification	10665	1413	32601

the data with K_d less than 30 as positive samples and the remaining samples as negative samples, thus using the BindingDB dataset for the binary classification task.

In this study, the dataset was divided by cold start instead of random division. The division into cold drug, cold protein, and cold pair was implemented by DeepPurpose software package [39]. Cold drug means that the drugs in the test set do not appear in the training set, while all the proteins appear in both sets. This experimental setup can be useful when one needs to identify potential proteins that may interact with a newly discovered compound. Cold protein means that none of the proteins in the test set appear in the training set, while drugs appear in both sets, and the significance of this split is enhancement of the ability to recommend potential drugs for new proteins for complex diseases. Cold pair means that neither the drugs nor the proteins in the test set appear in the training set, which can investigate whether the new compounds can serve as potential drugs for new protein targets, but this experimental setup is also most challenging.

3. Experiments

3.1. Evaluation metrics

We use Mean Squared Error (MSE) and Concordance Index (CI) [40] to evaluate the performance of the model on the regression task. MSE is a common metric that measures the difference between predicted values and actual values. The smaller the MSE, the closer the predicted values are to the true values, indicating better model performance. The formula for calculating the MSE can be found in Equation (8). The CI value is used to evaluate the model's predictive ability and is defined as the proportion of pairs of labels where the predicted and actual results are consistent. This evaluation metric is widely used in other DTA prediction methods, and a larger CI value indicates stronger predictive ability. The formula for CI is defined as:

$$CI = \frac{1}{z} \sum_{d_x > d_y} h(b_x - b_y) \quad (10)$$

where b_x is the predicted value of the larger binding affinity d_x , and b_y is the predicted value of the smaller binding affinity d_y . z is a standardization constant, and $h(x)$ is a step function as shown below:

$$h(x) = \begin{cases} 1, & x > 0 \\ 0.5, & x = 0 \\ 0, & x < 0 \end{cases} \quad (11)$$

And when the problem is considered as a classification task, the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) is used as an evaluation metric.

3.2. Experimental settings

All experiments in this work were carried out on a machine with a 32 GB T V100 card and Intel Xeon 4210 (2.4G, 10C) CPU. The model was implemented using Python 3.6 and PyTorch 1.8.2 (long-time support version) and trained using the Adam optimizer with a fixed learning rate of 0.0003 and a batch size of 128. The number of training epochs was set to 300 for the Davis dataset and 700 for the KIBA dataset. The hyperparameters of the experimental model were determined using a grid search, and the search range and selected values of the hyperparameters are shown in Table 2.

3.3. The performance of ColdDTA

We compared the results of ColdDTA with other prediction methods, including traditional machine learning models: Support Vector Machine (SVM), Random Forest (RF), and deep learning models: DeepDTA [7],

Table 2

Search range and selected values of hyperparameters for ColdDTA.

Hyperparameter	Search range	Selected value
Learning rate	$[3 \times 10^{-2}, 3 \times 10^{-3}, 3 \times 10^{-4}, 3 \times 10^{-5}]$	3×10^{-3}
Batch size	[32, 64, 128, 256, 512]	128
Drug feature extraction module output dimension	[32, 64, 128, 256, 512]	128
Protein feature extraction module output dimension	[32, 64, 128, 256, 512]	128
Number of neurons in the hidden layer of the drug-target fusion module	[32, 64, 128, 256, 512]	32

TransformerCPI [17], and MgraphDTA [12]. RF has shown better performance than deep learning methods in some DTA tasks [41], so a comparison with traditional machine learning methods is necessary. DeepDTA applied CNNs to SMILES strings and protein sequences to extract local residue patterns. When we transformed the problem into a classification task, a Sigmoid activation function was added at the end of DeepDTA to convert it into a binary classification problem and a hyperparameter search was performed. TransformerCPI was originally designed for binary classification tasks, and to enable the comparison, we replaced its classifier and loss function to adapt it to the regression task. MgraphDTA built a super-deep GNN with more graph convolutional layers to capture both the local and global structure of compounds, and gained more performance as a result. Consistent with previous work [20], the experiment employed a 5-fold cross-validation method, in which all data were evenly partitioned into five parts, with one part used as the test set and the remaining four parts used for training. This resulted in five different schemes of the dataset, with the average score being recorded as the final performance metric. The amount of data contained in each fold is shown in Table 3. Since the cold start data splitting method needs to ensure that drugs or proteins from the test set do not appear in the training set, the amount of data contained in each fold is not exactly equal, and in cold pair, this splitting method leads to a less total number of data. Meanwhile, we repeated the experiment five times, each time choosing a different random seed. In order to ensure the fairness of the experiments, the parameters of all the methods are either adopted from the original papers or determined through experiments, and the experiments are conducted on the same training and test sets, but due to the randomness of the dataset splitting process, this may lead to different experimental results from other papers.

Table 4 shows the MSE and CI results of ColdDTA with various baselines on the Davis dataset, with standard deviations in parentheses. Also, in Table 5, we compared ColdDTA with other baseline models on the KIBA dataset. Our model achieved the best predictive performance on both benchmark datasets under the cold-start experiment. It is worth noting that in the cold drug experiment of the KIBA dataset, the RF method performed better than other State-Of-The-Art (SOTA) methods, but due to data augmentation strategy and drug-target fusion module, ColdDTA outperforms RF in the consistency index. Meanwhile, to show

Table 3

Amount of data per Fold in different split methods.

Datasets		Fold1	Fold2	Fold3	Fold4	Fold5	Total
Davis	Cold drug	6188	6188	5746	5746	6188	30056
	Cold protein	5304	6596	5780	5576	6800	30056
	Cold pair	1358	1235	1302	1014	1106	6015
KIBA	Cold drug	23826	25183	24240	23072	21933	118254
	Cold protein	23996	21351	25458	20024	27425	118254
	Cold pair	4838	4319	4831	4696	5010	23694
BindingDB	Cold drug	6039	7504	5730	8536	4792	32601
	Cold protein	5844	5074	6562	6854	8267	32601
	Cold pair	1129	1380	1270	1248	1319	6346

that the differences between ColdDTA and other methods are statistically significant, we applied the Wilcoxon test [42]. The experimental results show that ColdDTA has a p-value > 0.05 compared to RF on the KIBA cold drug task, but ColdDTA significantly outperform all other methods at the significance level of 0.05 on the other tasks of the Davis and KIBA datasets.

To visualize the performance improvement of ColdDTA, we plotted the CI values from Tables 4 and 5 as Fig. 5, which contains the best results of the traditional machine learning method and the best results of the deep learning method. It can be observed that our method achieved significant performance improvements on multiple tasks, surpassing the best results by 0.033, 0.017 and 0.009 in the Cold drug, Cold protein, and Cold pair tasks of Davis, respectively. Similarly, for the cold drug, cold protein and cold pair tasks of the KIBA dataset, ColdDTA achieves performance improvements over the best-performing method by 0.003, 0.031 and 0.020 performance gains. Notably, in the KIBA cold drug task, our method outperformed the best result of the deep learning method by 0.039. The improvement in predictive performance is due to our data augmentation strategy that generates more drug-target pairs, while the attention-based drug-target fusion module allows the two objects to fuse at a mathematically, such that it better captures information about interactions between drugs and proteins.

Furthermore, to investigate the performance of the model in handling the classification task, we compared ColdDTA with the baseline models on the BindingDB dataset. The results are shown in Table 6. We are able to see that ColdDTA is competitive with all five baseline models on the classification task.

3.4. Ablation study

An ablation study was performed on the Davis dataset in order to validate the effectiveness of the proposed data augmentation strategy and the drug-target fusion module. We used the following settings:

1. - Drug-target fusion: We replaced the drug-target fusion module with a simple concatenation operation.
2. - Data augmentation: We further do not use data augmentation, and at this point, the model does not use any of the methods proposed in this work.

The experimental results are shown in Table 7, from which we can see that both attention-based feature fusion and data augmentation strategy contribute to the model's predictive performance.

Additionally, we separately investigated the impact of data augmentation on model prediction performance, and this part of the experiments all used the drug-target fusion module. The optimal parameters for using data augmentation in the benchmark datasets are shown in Table 8, and these parameters were obtained through grid search due to the limited range of hyperparameter search space, and the search range for removing subgraph ratios is [0.1, 0.2, 0.3, 0.4], and the search range using data augmentation times is [1, 2, 3, 4, 5]. The effect of whether or not to use data augmentation on prediction performance is shown in Fig. 6, where the length of the error bars represents the

Table 4

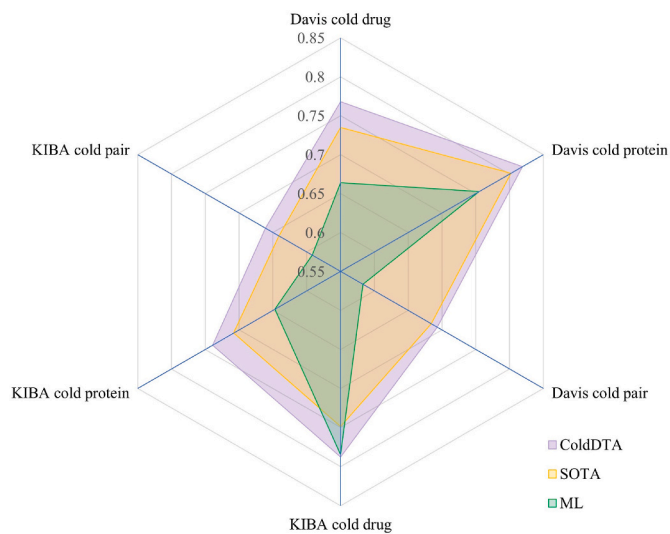
Performance comparison results with other methods on the Davis dataset (five random runs).

Methods	Cold drug		Cold protein		Cold pair	
	MSE	CI	MSE	CI	MSE	CI
SVM	0.722(± 0.016)	0.663(± 0.012)	0.849(± 0.010)	0.650(± 0.018)	0.925(± 0.009)	0.546(± 0.008)
RF	0.719(± 0.013)	0.664(± 0.012)	0.609(± 0.004)	0.755(± 0.002)	0.877(± 0.010)	0.583(± 0.007)
DeepDTA	1.078(± 0.023)	0.684(± 0.020)	0.592(± 0.043)	0.769(± 0.015)	0.802(± 0.032)	0.646(± 0.031)
TransformerCPI	0.867(± 0.027)	0.636(± 0.026)	0.483(± 0.022)	0.776(± 0.006)	0.891(± 0.038)	0.522(± 0.034)
MgraphDTA	0.573(± 0.011)	0.735(± 0.013)	0.410(± 0.014)	0.802(± 0.010)	0.707(± 0.018)	0.684(± 0.017)
ColdDTA(ours)	0.549(± 0.008)	0.768(± 0.009)	0.393(± 0.005)	0.819(± 0.003)	0.678(± 0.014)	0.693(± 0.014)

Table 5

Performance comparison results with other methods on the KIBA dataset (five random runs).

Methods	Cold drug		Cold protein		Cold pair	
	MSE	CI	MSE	CI	MSE	CI
SVM	0.639(± 0.013)	0.711(± 0.006)	1.097(± 0.015)	0.596(± 0.006)	0.750(± 0.016)	0.564(± 0.015)
RF	0.348(± 0.002)	0.785(± 0.001)	0.554(± 0.003)	0.647(± 0.002)	0.633(± 0.017)	0.592(± 0.026)
DeepDTA	0.570(± 0.025)	0.701(± 0.022)	0.438(± 0.020)	0.728(± 0.016)	0.677(± 0.042)	0.627(± 0.017)
TransformerCPI	1.293(± 0.023)	0.686(± 0.011)	1.132(± 0.021)	0.627(± 0.019)	1.292(± 0.024)	0.587(± 0.021)
MgraphDTA	0.414(± 0.013)	0.749(± 0.008)	0.440(± 0.017)	0.708(± 0.013)	0.589(± 0.013)	0.641(± 0.024)
ColdDTA(ours)	0.380(± 0.018)	0.788(± 0.009)	0.372(± 0.015)	0.739(± 0.011)	0.540(± 0.010)	0.661(± 0.019)

**Fig. 5.** CI results of ColdDTA compared with SOTA and traditional machine learning methods on public datasets.

standard deviation. From Fig. 6, we can see that using the data augmentation strategy proposed in this research can effectively enhance the model's predictive performance in the cold drug and cold pair tasks of the two public datasets. Specifically, when using the drug-target fusion module without the data augmentation strategy, the CI values for Davis cold drug, Davis cold pair, KIBA cold drug, KIBA cold pair were 0.757 ± 0.012 , 0.684 ± 0.018 , 0.776 ± 0.013 , 0.653 ± 0.023 , respectively. After using data augmentation, all CI values were significantly improved and the corresponding MSE values were reduced. But for the cold protein task, our strategy did not have a significant effect, and in the KIBA dataset, it even had a slightly negative impact, with the CI becoming 0.739 ± 0.011 . This study demonstrated that the data augmentation method is effective for the DTA regression task, and selecting a reasonable method can effectively improve the model's generalization ability. Removing subgraphs to form data pairs with unchanged binding affinity between the drug and the original target may seem to defy common biological sense, but in fact, multiple data augmentations to generate different subgraphs allow the training set to

Table 8

Optimal hyperparameters for data augmentation in each dataset.

Hyper-parameter	Davis	KIBA	BindingDB
Removed subgraph ratio	0.2	0.1	0.2
Times of data augmentation	4	4	4

Table 6

Performance comparison results with other methods on the binary classification dataset BindingDB (shown avg. ROC-AUC of five random runs).

Settings	SVM	RF	DeepDTA	TransformerCPI	MgraphDTA	ColdDTA(ours)
Cold drug	0.707(± 0.008)	0.763(± 0.009)	0.830(± 0.008)	0.802(± 0.008)	0.858(± 0.005)	0.875(± 0.003)
Cold protein	0.580(± 0.006)	0.613(± 0.003)	0.818(± 0.005)	0.695(± 0.010)	0.821(± 0.007)	0.840(± 0.005)
Cold pair	0.517(± 0.013)	0.560(± 0.016)	0.649(± 0.017)	0.740(± 0.019)	0.741(± 0.021)	0.757(± 0.014)

Table 7

Ablation study (five random runs).

Set up	Cold drug		Cold protein		Cold pair	
	MSE	CI	MSE	CI	MSE	CI
ColdDTA	0.549(± 0.008)	0.768(± 0.009)	0.393(± 0.005)	0.819(± 0.003)	0.678(± 0.014)	0.693(± 0.014)
- drug-target fusion	0.570(± 0.014)	0.761(± 0.013)	0.412(± 0.015)	0.807(± 0.002)	0.705(± 0.017)	0.681(± 0.019)
- data augmentation	0.582(± 0.013)	0.729(± 0.013)	0.417(± 0.015)	0.800(± 0.003)	0.731(± 0.024)	0.672(± 0.021)

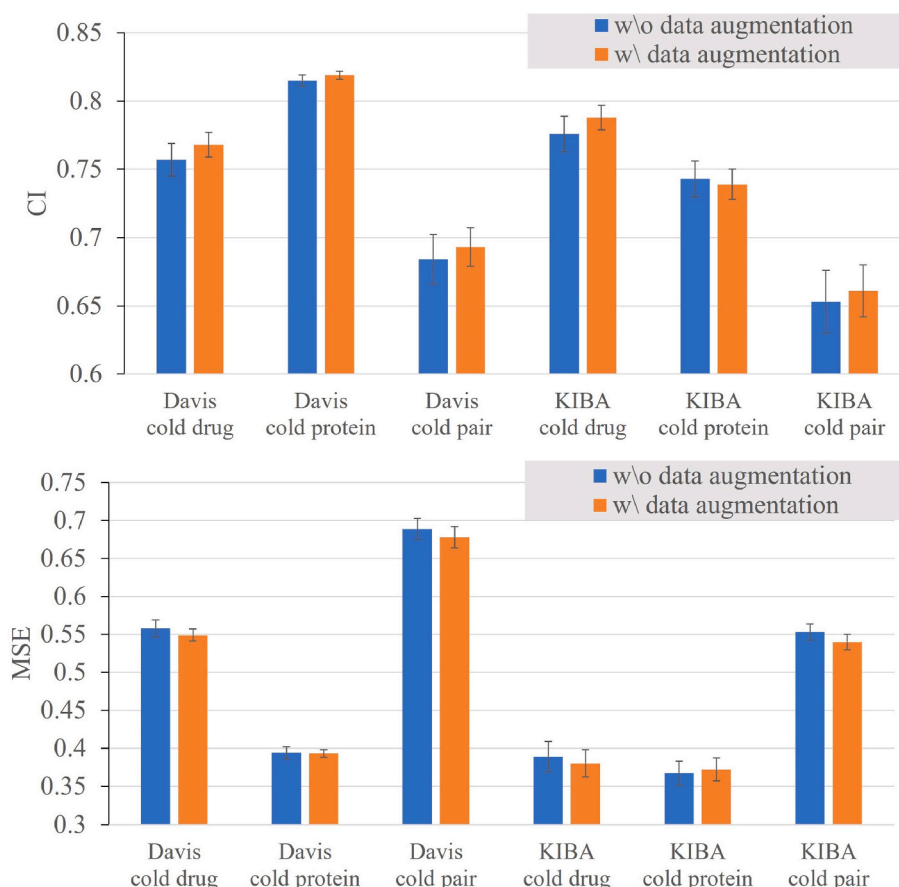


Fig. 6. Performance of the model with or without data augmentation (five random runs).

implicitly contain the full molecular map information, while the deep learning model is better able to learn key information from the remaining subgraphs, as our experiments demonstrate. At the same time, one should also realize that the use of a completely randomized data augmentation strategy may produce unphysical samples in a few cases, similar to many data augmentation methods in computer vision [42,43], and we will try to address this issue in our future work.

We also experimentally demonstrated the compatibility of our proposed data augmentation strategy, where we select Graph Attention Network (GAT), Graph Isomorphism Network (GIN), and Graph Convolutional Network (GCN) as feature encoders for drugs. The experiments were performed on the cold drug split of Davis since our strategy works best for cold drug and the training time is shorter compared to the KIBA dataset. The experimental results are shown in Table 9, which shows that our data augmentation method is well-compatible with various GNN networks.

3.5. Model interpretation

The black box problem of deep learning has always been

Table 9
Compatibility to other GNN methods (five random runs).

Methods	MSE	CI
GAT	0.700(±0.053)	0.673(±0.013)
GAT + Aug	0.645(±0.049)	0.712(±0.011)
GIN	0.714(±0.047)	0.675(±0.009)
GIN + Aug	0.683(±0.045)	0.694(±0.006)
GCN	0.704(±0.057)	0.679(±0.033)
GCN + Aug	0.684(±0.053)	0.695(±0.031)

* Aug means data augmentation.

controversial. Although people can see the results of the model's output, it is difficult to explain why the model made such judgments. This lack of interpretability limits the application of deep learning methods in drug discovery.

To demonstrate the interpretability of ColdDTA, we conducted a case study on the crystal structure (PDB: 3MDZ) of the protein Carbonic anhydrase 7 that binds with ethoxzolamide. We input the drug and its amino acid sequence into the model, obtained the binding affinity results, and calculated the weights of each atom with each amino acid. Finally, we mapped the weights for visualization of importance. We used Grad-AAM [12] and RDkit [44] to display the weights of the compound, and Chimera for protein visualization [45]. The results are shown in Fig. 7. The red parts in the protein represent amino acids with higher weights, with a total of 7 amino acids obtaining weights greater than 0.9, including GLN 94 and HIS 96 with weights greater than 0.98. Most of the high-weight amino acid residues overlap with the actual interaction sites, although we also noticed that some red-marked amino acids seem to have long range interaction to the compound. In the corresponding compound, the red-marked parts indicate atoms with attention. The ligand functional groups of attention have high similarity to the results observed in 3MDZ. The above results suggest that our model can provide mechanistic insights to researchers, but we would also like to point out that the interpretability of the model may not be statistically confirmed.

4. Conclusion

In this study, we proposed an end-to-end training method using data augmentation and attention-based feature fusion for predicting drug-target binding affinity: ColdDTA. It removes a fixed ratio of drug molecule graphs in the training phase to form new data pairs with the

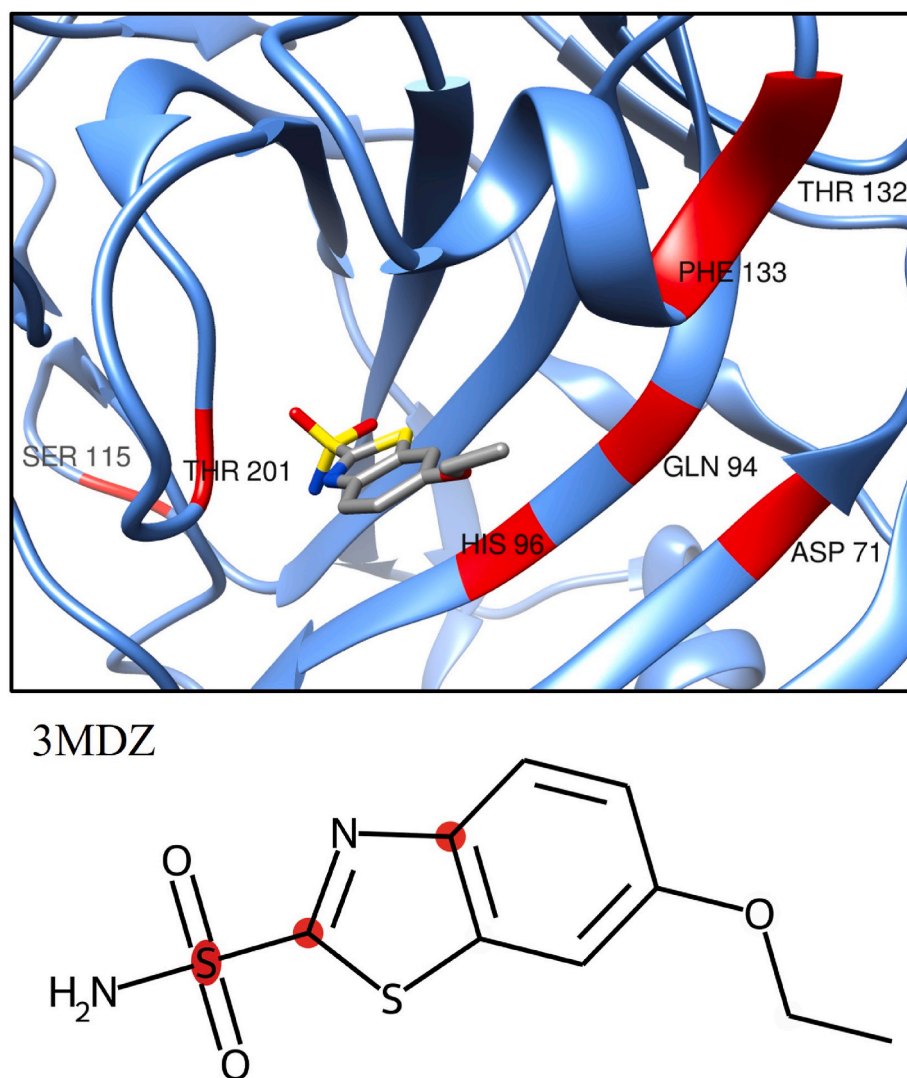


Fig. 7. Visualization of the importance of drugs and targets.

original targets, and this data augmentation can effectively improve the generalization ability of the model. Meanwhile, the model uses a new attention-based feature fusion method to better interact compound with protein information, thus improving the model prediction performance. Experimental results show that ColdDTA significantly improved performance on the benchmark dataset compared with SOTA baseline and traditional machine learning methods. The effectiveness of the proposed approach is also verified by ablation experiments. In addition, we demonstrated the interpretability of the model through the visualization of weights.

Declaration of competing interest

None declared.

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Foshan National Hi-tech Industrial Development Zone. The code is available at <https://github.com/doragu/coldDTA>.

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